INCH-POUND

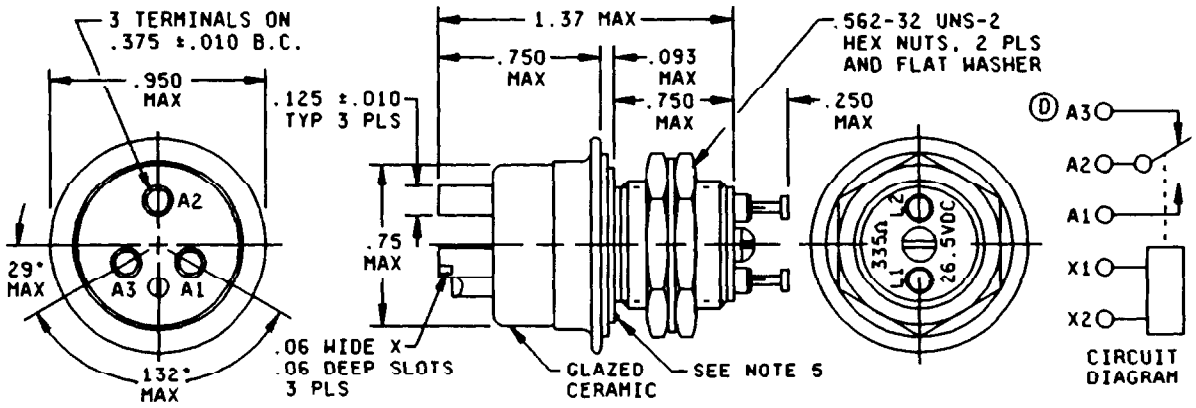
MIL-R-83725/5D(USAF)
15 April 1994
SUPERSEDING
MIL-R-83725/5C(USAF)
17 January 1992

MILITARY SPECIFICATION SHEET

RELAYS, VACUUM, SPDT, 18 AMPERES, RMS, 3.5 KILOVOLTS PEAK (CARRY ONLY)

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-83725.



Inches	mm
.020	0.51
.030	0.76
.060	1.52
.083	1.65
.125	2.36
.250	6.35
.520	13.21
.750	19.05
.950	24.13
1.37	34.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Part or Identifying Number (PIN) shall be permanently marked on the relay envelop.
5. A .030 X .060 X .020 maximum rectangular index slot shall be used to accommodate the index pin.

D FIGURE 1. Relay outline and dimensions.

D denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:25Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

CONTACT DATA:

Configuration: SPDT.

Arrangement: One form C.

Load ratings:

Resistive: Carry only:

18 amperes dc or 60 Hz rms.
14 amperes rms, 2.5 MHz.
9 amperes rms, 16 MHz.
7 amperes rms, 32 MHz.

Rated operating voltage:

3.5 kV dc: 60 Hz or dc.
2.5 kV dc: 2.5 MHz.
2.0 kV dc: 16 MHz.
1.5 kV dc: 32 MHz.

Contact resistance:

Rated life:

Before: .01 ohm, maximum.
During: .100 ohm, maximum.
After: .100 ohm, maximum.

Contact bounce: Not applicable.

Capacitance:

2 picofarads, maximum across open contacts.
2.5 picofarads, maximum between open contacts and ground.

(D) COIL DATA:

Duty rating: Continuous.

Maximum voltage: See table I.

Nominal voltage: See table I.

Pickup voltage: See table I.

Dropout voltage: See table I.

Coil resistance: see table I.

Operate time: 8 milliseconds, maximum over temperature range (or measured using 18.8 V dc coil voltage at 25°C).

Release time: 8 milliseconds, maximum over temperature range (or measured using 18.8 V dc coil voltage at 25°C).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:25Z
Check the source to verify that this is the current version before use.

ELECTRICAL DATA:

Insulation resistance: 1,000 megohms, minimum, except the resistance between coil and housing shall be 500 megohms or greater.

Dielectric withstanding voltage:

At atmospheric pressure:

Between all mated contacts in the open position: 5 kV dc peak.
Between high voltage terminals and housing: 5 kV dc peak.
Between coil and housing: 500 V rms, 60 Hz.

(D) At reduced barometric pressure:

In accordance with MIL-STD-202, method 105, test condition C. Relays shall be mounted by normal mounting means. Points of application: Same as "at atmospheric pressure" except test voltage applied shall be 350 volts rms with other terminals grounded. Following the test, relays shall be examined for evidence of arcing, flashover, insulation breakdown, and damage.

ENVIRONMENTAL DATA:

Temperature range: -55°C to +125°C.

Vibration: MIL-STD-202, method 204, test condition C (10 to 55 Hz, .06 inch (1.5 mm) double amplitude 55 to 2,000 Hz, 10 g's).

Shock: MIL-STD-202, method 213, test condition A, 50 g's, 11 milliseconds half sine.

PHYSICAL:

Terminal strength: 5 pounds pull.

Dimensions and configuration: See figure 1.

Termination: Solder terminal.

Weight: 1 ounce, maximum.

Marking: Applicable (circuit diagram need not be marked on relay).

(D) TABLE I. Operating characteristics. 1/

Dash number	Maximum voltage	Nominal voltage	Pickup voltage	Dropout voltage	Coil resistance	Pickup voltage over temperature range	Dropout voltage over temperature range
-001	32 V dc	26.5 V dc	14 V dc	1.5 - 7 V dc	335 ohms ±10%	22 V dc	1 - 10 V dc

1/ At +25°C.

LIFE TEST REQUIREMENTS:

Mechanical cycling: 1 million cycles, 2 sample units (cycling rate, 36,000 per hour maximum; contact current shall not exceed 10 milliamperes).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:25Z
Check the source to verify that this is the current version before use.

QUALITY ASSURANCE:

(D) Dielectric withstanding voltage (for group A testing):

Tests to be conducted at atmospheric pressure rating only.

Duration of application: 5 to 10 seconds at a 10 percent increase in the dielectric withstanding voltage.

PIN: M83725/5 001.

CONCLUDING MATERIAL

Review activity:
Air Force - 99

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F737-01)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:25Z
Check the source to verify that this is the current version before use.